CISCO MCP SUITE · CUSTOMER-FACING TEST REPORT

Security Intelligence — Threat-Feed Blocking

C7 — an FMC Security Intelligence blocklist drops a listed destination **before** the access-control rules run (pre-ACL), while a control destination on the same allow rule stays reachable

RUN DATE	TESTER	DESIGN	PLAN
2026-07-17	**testing-agent (live)**	**SD-Access ISE Integration**	**SI-001…006**

VERDICT PASS-with-caveat

5 PASS · 1 partial · 0 FAIL. Adding the **host-splunk** Host object (198.18.128.51) to the ACP's Security-Intelligence blocklist and deploying flipped **HOST1 → 198.18.128.51 from 0% → 100% loss** while the control **HOST1 → 198.18.134.35 stayed 0% loss** — a provably **pre-ACL** drop. Removing the entry restored 0% loss. The single caveat is the documented **D13** telemetry gap: the SI *block* event reaches the FMC event viewer but **not Splunk** (scored partial, not a C7 failure). Read-only on built config; one reversible SI round-trip, fully undone and re-verified.

CML lab "SDA-Fabric" · 77dd2fde-1fda-4cc9-9b29-48ff98bd1395 · FMC 10.0.1 · FTD 10.0.0 (Snort 3) · ISE 3.5.0.527 · cat9000v IOS-XE 17.18

Sibling to the C2/C6 rapid-threat-containment reports (report-ANC.pdf, report-C6-IPS-RTC.pdf) — same lab, a different firewall control (pre-ACL threat-feed blocking).

1. Executive summary

Live, independent re-verification of roadmap item **C7 — Security Intelligence (SI) threat-feed blocking** (`SI-001…006`) on the **SD-Access ISE Integration** lab. An FMC **Security Intelligence** blocklist entry makes the FTD drop traffic to a listed destination **pre-ACL** (ahead of the access-control rules), while a control destination on the *same* allow rule stays reachable. Executed as a **single reversible round-trip** — add the block → verify → remove it — against the live FMC/FTD and the SDA fabric.

> **The flip, observed end to end.** From baseline (SI `networks.blocklist` = only **Global Block List**; HOST1 → 198.18.128.51 **0% loss**), the **host-splunk** Host object (198.18.128.51 — an otherwise ACL-allowed, reachable stand-in "threat") was appended to the SI blocklist over the FMC REST API and deployed. After deploy: **HOST1 (alice, 172.16.10.50) → 198.18.128.51 = 100% loss (blocked)** while the control **HOST1 → 198.18.134.35 (ISE) = 0% loss (reachable)** — only the listed dst is dropped, and because it was ACL-allowed at baseline the drop is provably **pre-ACL**. Removing the host and redeploying **restored 0% loss**.

Result: 5 PASS · 1 partial · 0 FAIL. The partial is the documented **D13 telemetry gap** (SI-005): the SI *block* security event is logged to the FMC event viewer but does **not** reach Splunk. This run additionally found that connection events (`430002/430003`) **do** now land in Splunk (an improvement since C6), but the SI-block security event specifically still does not — a precise narrowing of D13. Scored **partial / known-gap**, not a hard FAIL of C7 (the capability under test — the pre-ACL block — passed). No built configuration was modified.

2. Scope & systems under test

Test plan: Lab Designs / security-intelligence-blocking (`SI-001…006`). **Design:** C7 on the SD-Access ISE Integration lab; context module `Custom Designs/SD-Access ISE Integration/modules/fmc-security-intelligence.md` .

What the lab is (plain English). A virtual **Cisco SD-Access campus fabric** — a LISP/VXLAN overlay with a collapsed control-plane/border node **BORDER-CP** and a fabric edge **EDGE1**, fused to the outside world by **FUSION-R1** — carrying a full identity + security stack. **Cisco ISE 3.5** is the RADIUS/TrustSec/pxGrid policy server; **Microsoft AD** (`mitchcloud.lab`) is the external identity store; a **Cisco Secure Firewall (FMC + FTD)** sits inline at the fabric's external edge (traffic to outside destinations is policy-routed from the fusion through the FTD); **Splunk** is the telemetry sink. Employee endpoint **alice / HOST1** authenticates with 802.1X, lands in the **Employees** security group, and gets normal fabric access.

What C7 adds. A firewall control that acts **ahead of** the access-control rules. **Security Intelligence** matches a destination against block lists (reputation / threat feeds) at the very front of the packet path; a match is dropped as a *Security-Related Connection → Block → IP Block*, and the allow rule that would otherwise permit it never runs. The FMC REST API cannot create custom SI *list* objects (GUI-only), but it **can** add a **Host/Network object** to the ACP's SI `networks.blocklist` — that object becomes the "custom threat list" entry.

```
HOST1 (alice 172.16.10.50) → EDGE1 → BORDER-CP → FUSION-R1 (PBR) → FTDv
  FTDv Security Intelligence: dst on networks.blocklist?
     yes (host-splunk 198.18.128.51)  → DROP pre-ACL          → 100% loss
     no  (ISE 198.18.134.35, control) → AC rule Allow (normal) →   0% loss
```

Component versions verified live this run

Component	Version
Cisco Secure Firewall Management Center (FMC)	**10.0.1 (build 1)**
Cisco Secure Firewall Threat Defense (FTD)	**10.0.0** — Snort 3
Cisco ISE (control dst / identity)	**3.5.0.527**
Fabric edge / NAD (EDGE1)	**cat9000v IOS-XE 17.18**
SD-Access fabric	**LISP/VXLAN, Closed Authentication**

3. Test environment

Lab: CML "SDA-Fabric" (`77dd2fde-1fda-4cc9-9b29-48ff98bd1395`) — STARTED / converged, all nodes BOOTED. IPs are **lab-specific**.

Hostname	Role / node-type	Mgmt IP	Data IP(s)	VRF / VLAN
HOST1 (`alice`)	fabric endpoint (test source) · alpine	—	**172.16.10.50** · MAC 52:54:00:03:0B:0D	CAMPUS_VN
EDGE1	fabric edge / NAD · cat9000v	198.18.128.73	fabric-attached	CAMPUS_VN
BORDER-CP	fabric control-plane + border · cat9000v	198.18.128.72	LISP map-server / L3 handoff	CAMPUS_VN

Hostname	Role / node-type	Mgmt IP	Data IP(s)	VRF / VLAN
FUSION-R1	fusion router (PBR → FTD inside) · cat8000v	198.18.128.71	inside handoff	global
FTDv	Secure Firewall TD — **SI enforcer** · ftdv	198.18.128.81	data/syslog src **198.18.128.82**	routed
FMCv	Mgmt Center — owns SDA-ACP + SI policy · fmcv	198.18.128.80	—	—
host-splunk	**listed "threat"** (test) + Splunk SIEM · splunk	—	**198.18.128.51**	—
ISE 3.5 (`ise35`)	**control dst** (same allow rule, never listed)	**198.18.134.35**	—	external VM

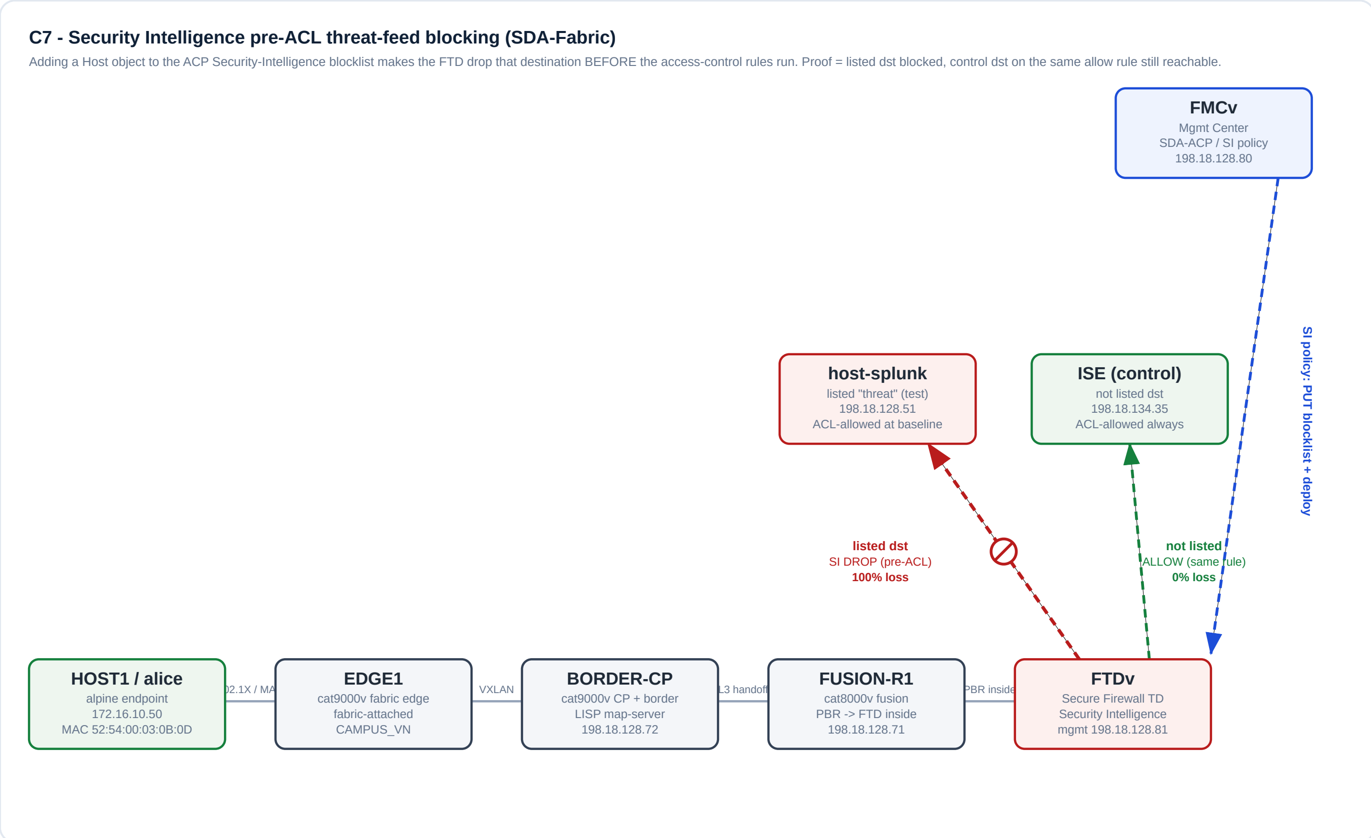


Figure 1 — C7 Security-Intelligence pre-ACL block chain. The FTD evaluates the destination against the SI blocklist ahead of the AC rules: the listed dst (host-splunk, red) is dropped pre-ACL (100% loss) while the control dst (ISE, green) on the same allow rule stays reachable (0% loss). FMCv pushes the SI policy (PUT + deploy).

4. Results — automated gate (MCP servers)

Not applicable to this run. C7 acceptance is **manual-live** end-to-end against the live FMC/FTD + SDA fabric + Splunk; there is no offline CI gate for a lab-design capability. The underlying MCP tool health (firepower-mcp, cml-mcp, splunk-mcp) is covered by those servers' own plans and the 2026-07-15 automated-gate report. All live surfaces answered this run: FMC token auth + config/deployment API OK; FTDv MANAGED/DEPLOYED; HOST1 pyATS console OK; Splunk management API + SPL search OK.

5. Results — lab-design acceptance

Manual-live, case-by-case. Round-trip: append **host-splunk (Host, 198.18.128.51)** to the SI `networks.blocklist` → deploy → verify → remove → deploy → verify.

ID	Objective	Result	Evidence
`SI-001`	Baseline (no SI block): listed + control dst reachable	PASS	GET SI policy → `networks.blocklist` = only **Global Block List** (SINetworkList). HOST1 eth0 172.16.10.50/24. HOST1 → **198.18.128.51 = 0% loss** (4/4, ~87ms); → **198.18.134.35 = 0% loss** (4/4, ~81ms). host-splunk reachable at baseline ⇒ it is ACL-allowed, so a later block must be pre-ACL.
`SI-002`	SI REST API limits are as documented	PASS	`POST /object/sinetworklists` → **HTTP 405** (custom SI lists GUI-only). `Global-Block-List` `metadata.readOnly.state=true`. SI blocklist rejects **NetworkGroup** ("not allowed in Security Intelligence Policy") → Host/Network only. host-splunk = **Host**, value 198.18.128.51.

ID	Objective	Result	Evidence
`SI-003`	Apply — add the Host object to the SI blocklist	PASS	GET → stripped `links` / `metadata` → appended `{network: host-splunk, type Host}` → **PUT 200**. Read-back `networks.blocklist` = [**Global Block List**, **host-splunk (Host)**]. Deploy task **4294973908** → **Deployed / SUCCEEDED**. `blocklistLogging.sendLogsToSyslogServer=true`.
`SI-004`	Verify pre-ACL block (listed blocked, control reachable)	PASS	HOST1 → **198.18.128.51 = 100% loss** (0/5) → SI-blocked. HOST1 → **198.18.134.35 = 0% loss** (5/5) → still reachable. Only the listed dst is dropped; it was ACL-allowed at baseline (SI-001) ⇒ the drop is **pre-ACL** (Security Intelligence, ahead of the AC rules).
`SI-005`	SI block event → Splunk (SIEM telemetry)	PARTIAL	Msgid histogram (`index=network host=198.18.128.82`) → LINA `%FTD-*` **plus** connection events **430002/430003** (16 each). Connection **allow** events for the control dst & pre-block allows to .51 **are** present (improvement vs C6-era “no 430xxx”). **But** the post-block blocked pings to 198.18.128.51 produced **zero** Splunk events, and **0** events with `AccessControlRuleAction: Block` / “Security Intelligence” / 430001 in 45 min → the **SI block security event does not reach the SIEM** (FMC event viewer only) — **known gap D13**.
`SI-006`	Revert — remove the SI block, restore baseline	PASS	GET → removed host-splunk → **PUT 200**; read-back = [**Global Block List**]. Deploy task **4294974162** → **Deployed**; pending devices **0**. HOST1 → **198.18.128.51 = 0% loss** (5/5) restored; → 198.18.134.35 = 0% loss. End-state `networks.blocklist` = [**Global Block List**] (baseline).

6. Summary statistics

Metric	Value
Cases executed	6
PASS	**5**
Partial (known gap)	**1** (SI-005 / D13)
FAIL / Skip / Unreachable	0 / 0 / 0
Reachability flip (HOST1 → host-splunk)	0% → **100%** → 0% loss
Control dst (HOST1 → ISE)	0% loss throughout (never listed)
SI blocklist end-state	**Global Block List only** (baseline)
Deployments	2 (apply + revert), both **Deployed / SUCCEEDED**
Built config modified	**None** (read-only + one reversible SI round-trip, undone + re-verified)

7. Observations & defects

Defects raised: none new. Five cases passed and the round-trip was fully reverted; the one partial is a **pre-existing, documented** telemetry gap (D13), carried as a follow-up rather than a C7 failure.

- **D13 narrowed by this run.** The C6 report recorded “no 430xxx in Splunk”. This run found `430002/430003` **connection** events *do* now land in Splunk (allow events for the control dst and pre-block traffic to .51). What is still missing is the **SI *block* security event** for the blocked dst — so D13 is more precisely: *the FTD security-event syslog stream (SI block / 430001 intrusion / block-action connection) is not delivered to Splunk, even though ordinary connection events are.*
- **FTDv health “yellow” (cosmetic).** The FTD reported `health=yellow` but was MANAGED/DEPLOYED and enforced the SI policy correctly (blocked the listed dst, passed the control).
- **The block is genuinely pre-ACL, not an ACL deny.** host-splunk pinged clean at baseline (SI-001) under the same allow rule that keeps the control dst reachable; only after adding it to the SI blocklist did it flip to 100% loss.

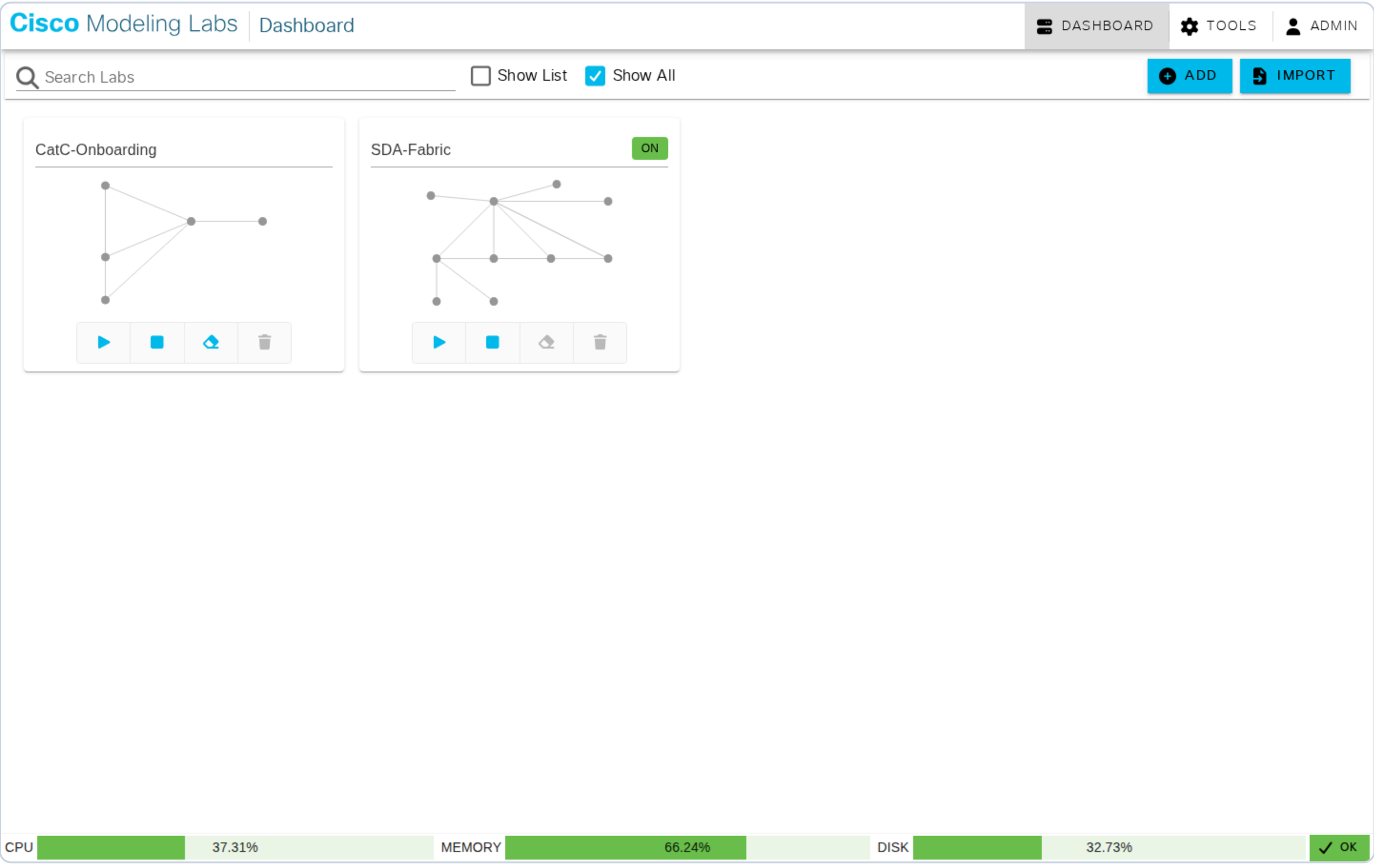
C7 API-limit gotchas (re-confirmed this run)

- **Custom SI list/feed objects are GUI-only:** `POST /object/sinetworklists` → **405**; the editable `Global-Block-List` is `readOnly` over REST. Add a **Host/Network object** to the ACP’s SI `networks.blocklist` instead — that object *is* the custom-threat entry.
- **SI blocklist rejects `NetworkGroup`** — it must be a **Host** or **Network** object.
- **`fmc_api_call` won’t take a JSON-object body** for the SI PUT → use **curl** (token via `generatetoken`, short TTL → re-auth per call); **strip `links` / `metadata`** from the GET body before PUT-ing it back.

Remediation brief — hand back (not applied here). Device group: FTDv (`37ffcab8-8126-11f1-a47b-d7a4b50be775`) + FMCv 198.18.128.80. **Owner:** firewall-engineer (+ splunk-engineer for verification). **Finding (D13):** FTD **security-event** syslog — SI blocks, `430001` intrusion, and block-action connection events — is not delivered to Splunk, though LINA `%FTD-*` and now ordinary `430002/430003` connection events are. The ACP SI `blocklistLogging` (`sendLogsToSyslogServer=true` , `sendLogsToEventViewer=true`) and IPS syslog toggles are already enabled; the security-event stream still does not arrive. **Change:** wire the FTD security-event syslog delivery path (FMC → Devices → Platform Settings → Syslog, plus the ACP Logging / Security-event & Intrusion settings), confirming the security-event syslog egresses an interface/route that reaches Splunk 198.18.128.51. **Acceptance check:** re-run the SI round-trip; Splunk `index=network host=198.18.128.82` shows a **Block-action / Security-Intelligence** connection event for the blocked dst within the block window. One fix lights up **C6 IPS + C7 SI + connection-block** events in the SIEM together.

8. Appendix

- **Figure 1** — labeled C7 SI-block chain (`topology-C7.svg` , embedded in §3).
- **CML canvas capture** (`c7-cml-canvas.png` , below) — ground-truth: SDA-Fabric ON, converged.
- **`results-C7.json`** — machine-collected raw results for all six cases.
- Raw evidence transcripts (FMC REST GET/PUT of the SI policy before/after, deployment task polls, HOST1 pings baseline → blocked → restored, Splunk msgid histogram + SI-event search) captured inline in §5 and below.



CML dashboard ground-truth — the “SDA-Fabric” lab is ON / converged during the run.

Key raw evidence

```
SI networks.blocklist :  [Global Block List]  ->  [Global Block List, host-splunk (Host)]  ->  [Global Block List]
FMC deploy            :  task 4294973908 SUCCEEDED (apply) ; task 4294974162 Deployed (revert), pending=0

HOST1 -> 198.18.128.51 (listed) :  0% loss  ->  100% loss  ->  0% loss (restored)
HOST1 -> 198.18.134.35 (control):  0% loss  ->    0% loss  ->  0% loss   (never listed)

POST /object/sinetworklists            -> HTTP 405   (custom SI lists GUI-only)
GET  Global-Block-List                 -> metadata.readOnly.state = true
SI blocklist + NetworkGroup            -> rejected ("not allowed in Security Intelligence Policy")

Splunk (index=network host=198.18.128.82):
  msgid histogram: 110002/111001/111004/111008/111010/199017/302010/305011/305012/771002/780005  +  430002/430003
  430002/430003 events observed = AccessControlRuleAction: Allow (control + pre-block traffic)
```

```
post-block blocked pings to 198.18.128.51                       -> 0 Splunk events
"AccessControlRuleAction: Block" / "Security Intelligence" / 430001 -> 0 events  (SI block event not in SIEM = D13)
```